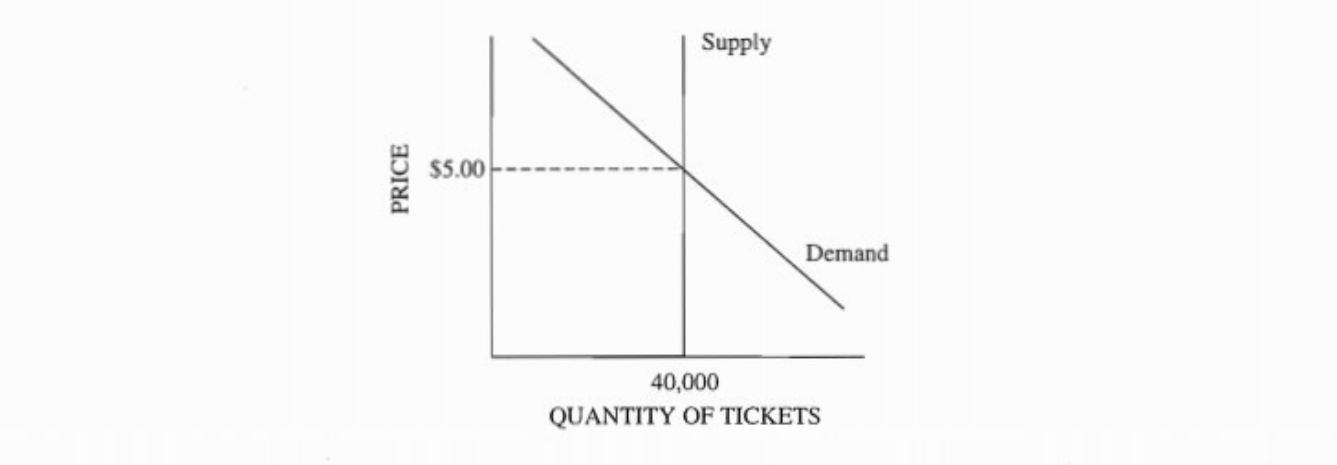
2.8 government intervention FRQ



The Toledo arena holds a maximum of 40,000 people, as indicated in the graph above. Each year the circus holds eight performances, all of which are sold out.

a. Analyze the effect on each of the following of the addition of a fantastic new death-defying trapeze act that increases the demand for tickets.

   i. The price of tickets

   ii. The quantity of tickets sold

b. The city of Toledo institutes an effective price ceiling on tickets. Explain where the price ceiling would be set. Explain the impact of the ceiling on each of each of the following.

   i. The quantity of tickets demanded

   ii. The quantity of tickets supplied

c. Will everyone who attends the circus pay the ceiling price set by the city of Toledo? Why or why not?

1. Respond to all parts of the question.

Assume that bread and butter are complementary goods. The government begins to subsidize the production of wheat, which is an input in the production of bread.

a. For each of the following markets, draw correctly labeled supply and demand graphs and show the effect of the subsidy on the equilibrium price and quantity in the short run.

   i. The wheat market

   ii. The bread market

   iii. The butter market

b. If the demand for bread is price elastic, how will total revenues for the bread producers change as a result of the government subsidy?

2.8 government intervention FRQ

2. Respond to all parts of the question.

Assume that the market for home security systems is perfectly competitive and currently in equilibrium.

a. Using a correctly labeled graph of supply and demand, show each of the following.

   i. The equilibrium price and quantity, labeled as P* and Q*, respectively

   ii. The area representing consumer surplus, labeled as CS

   iii. The area representing producer surplus, labeled as PS

b. Suppose that the government imposes an effective (binding) price ceiling. Redraw your graph in part (a), and label the ceiling price as P_2. Completely shade the area representing the sum of the consumer surplus and the producer surplus after the imposition of the price ceiling.

c. Suppose the demand for home security systems decreases and the price ceiling remains binding. Indicate what will happen to each of the following.

   i. Consumer surplus

   ii. Producer surplus

3. Respond to all parts of the question.

Directions: You are advised to spend the first 10 minutes reading all of the questions and planning your answers. You will then have 50 minutes to answer all three of the following questions. You may begin writing your responses before the reading period is over. It is suggested that you spend approximately half your time on the first question and divide the remaining time equally between the next two questions. Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. Use a pen with black or dark blue ink.

The markets for bananas, muffins, and coffee are interrelated, and each market is perfectly competitive.

a. *In the market for bananas, the equilibrium price is $1.00 per pound, and the equilibrium quantity is 1,000 pounds per week. Suppose the government imposes a price floor on bananas at $1.20* per pound, causing the quantity supplied to increase to 1,500 pounds per week.

   i. Would the price floor result in a shortage, a surplus, or neither? Explain.

   ii. Calculate the price elasticity of supply if the price increases from $1 to $1.20. Show your work.

   iii. Between $1 and $1.20, is the supply elastic, unit elastic, or inelastic? Explain.

b. Bananas are an input for muffins.

   i. Draw a correctly labeled graph of the market for muffins indicating the equilibrium price and quantity, labeled P_0 and Q_0, respectively.

2.8 government intervention FRQ

ii. On the graph drawn in part (b)(i), show the impact of an increase in the price of bananas on the muffin market, labeling the new equilibrium price and quantity P_1 and Q_1, respectively.

iii. On the same graph, completely shade the area that represents the **change** in the consumer surplus caused by the increase in the price of bananas.

c. In the market for coffee, the equilibrium price is $3.00 per cup and the equilibrium quantity is 100 cups per week. The cross-price elasticity of coffee with respect to muffins is -2.

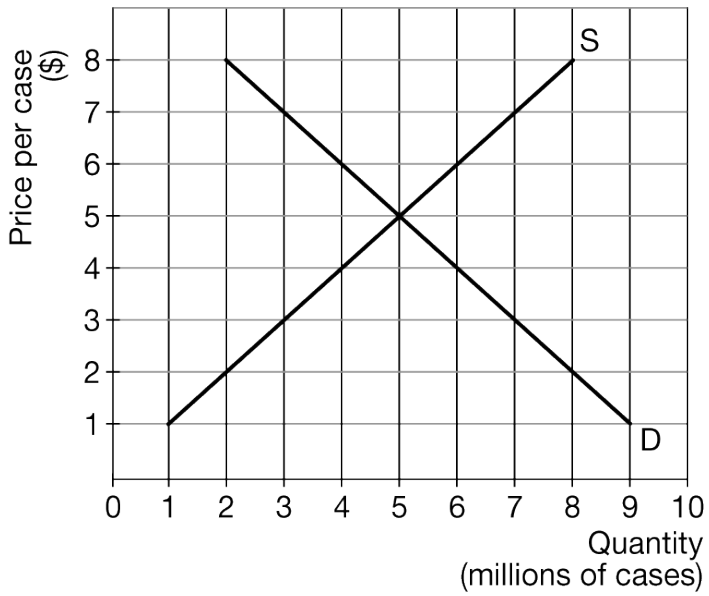
i. Are coffee and muffins normal goods, inferior goods, complementary goods, or substitute goods?

ii. Assume the supply of coffee is perfectly elastic. Using the equilibrium price and quantity given above, draw a correctly labeled graph for the coffee market, and show the impact of an increase in the price of muffins on the coffee market.

iii. Given the original quantity of 100 cups of coffee per week, if the increase in the price of muffins is 10%, calculate the new equilibrium quantity in the coffee market. Show your work.

4. Respond to all parts of the question.

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.



Assume that sugar-based soft drinks are produced in a market shown on the graph above. Answer the following questions based on the information given in the graph.

(a) To reduce the consumption of sugary soft drinks, suppose the government imposes a $2 per-unit sales tax on soft drinks.

(i) Will the price of soft drinks increase by the full amount of the per-unit sales tax? Explain.

(ii) Calculate the tax revenue the government can collect from the sale of soft drinks. Show your work.

2.8 government intervention FRQ

(iii) Will the consumer surplus increase, decrease, or stay the same after the per-unit sales tax?

(iv) Calculate the deadweight loss created by the per-unit sales tax. Show your work.

(b) Suppose that instead of imposing the per-unit sales tax, the government sets a price ceiling of $7. Identify the quantity of soft drinks that will be exchanged in the market as a result of the price ceiling. Explain.

5. Respond to all parts of the question.

In each part below, assume that the government imposes a per-unit sales tax and that the supply curve is upward-sloping.

a. In industry X, consumers buy the same quantity no matter what the price is.

   i. Using a correctly labeled graph, show what happens to the quantity sold when the tax is imposed.

   ii. How will the burden of the tax be distributed between buyers and sellers?

b. In industry Y, the market demand curve is perfectly elastic.

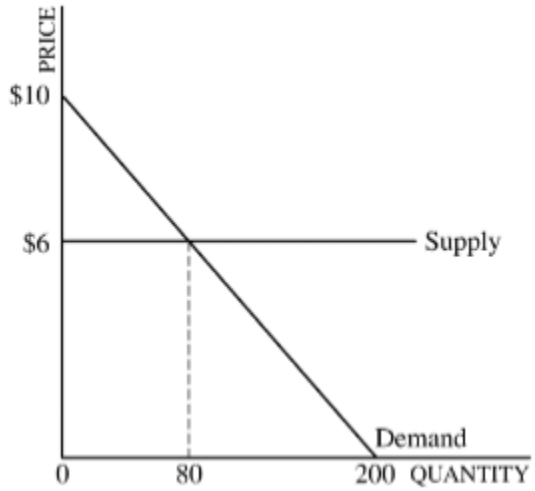
   i. Using a correctly labeled graph, show what happens to the price of the good that the consumers pay when the tax is imposed.

   ii. How will the burden of the tax be distributed between buyers and sellers?

c. In industry Z, the market demand curve is downward-sloping. Using a correctly labeled graph, shade the area that represents total tax revenues.

6. Respond to all parts of the question.

The market supply and demand for a product are shown in the diagram below.



2.8 government intervention FRQ

7. Respond to all parts of the question.

a. Is the price elasticity of supply less than one, equal to one, or greater than one? Explain.

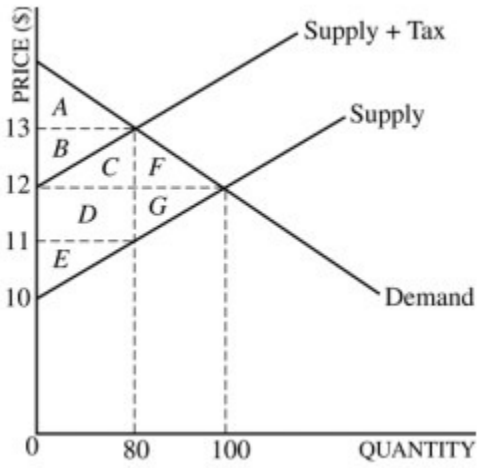
b. Calculate consumer surplus at the equilibrium price. Show your work.

c. Now suppose the government imposes a per-unit tax of $1 on producers.

i. What happens to total revenue received by producers after they pay the tax to the government? Explain.

ii. Will producer surplus increase, decrease, or stay the same?

iii. Will total surplus increase, decrease, or stay the same? Explain.



The graph above shows the market for a good that is subject to a per-unit tax. The letters in the graph represent the enclosed areas.

a. Using the labeling on the graph, identify each of the following.

i. The equilibrium price and quantity before the tax

ii. The area representing the consumer surplus before the tax

iii. The area representing the producer surplus before the tax

b. Assume that the tax is now imposed. Based on the graph, does the price paid by the buyers rise by the full amount of the tax? Explain.

c. Using the labeling on the graph, identify each of the following after the imposition of the tax.

i. The net price received by the sellers

ii. The amount of tax revenue

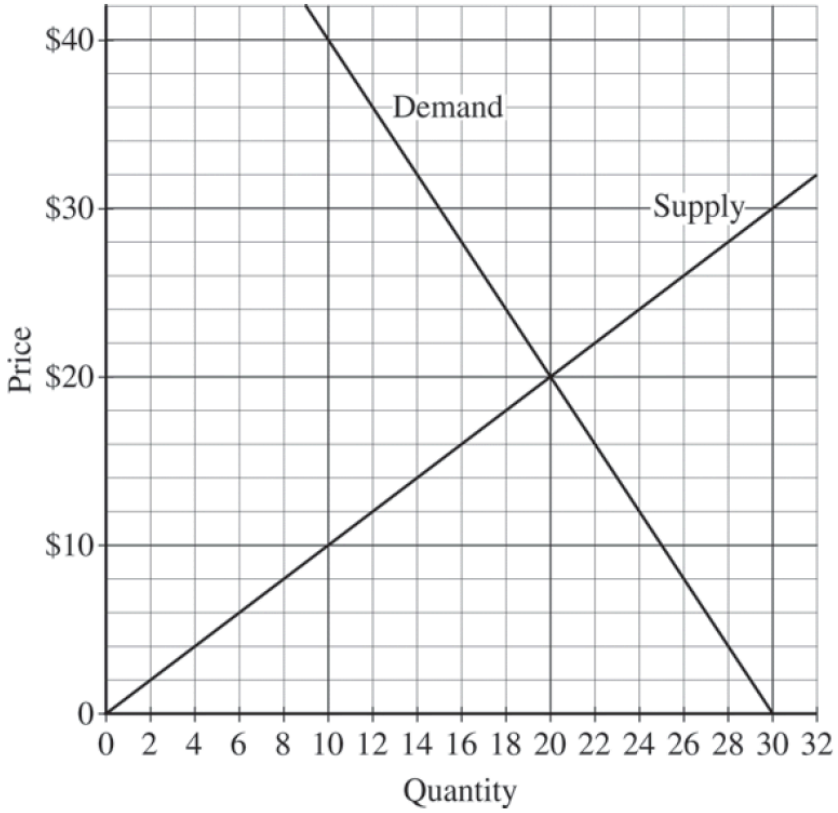
iii. The area representing the consumer surplus

2.8 government intervention FRQ

iv. The area representing the deadweight loss

8. Respond to all parts of the question.

The graph below shows the market for widgets. The government is considering intervening in this market.



a. Calculate the total producer surplus at the market equilibrium price and quantity. Show your work.

b. If the government imposes a price floor at $16, is there a shortage, a surplus, or neither? Explain.

c. If instead the government imposes a price ceiling at $12, is there a shortage, a surplus, or neither? Explain.

d. If instead the government restricts the market output to 10 units, calculate the deadweight loss. Show your work.

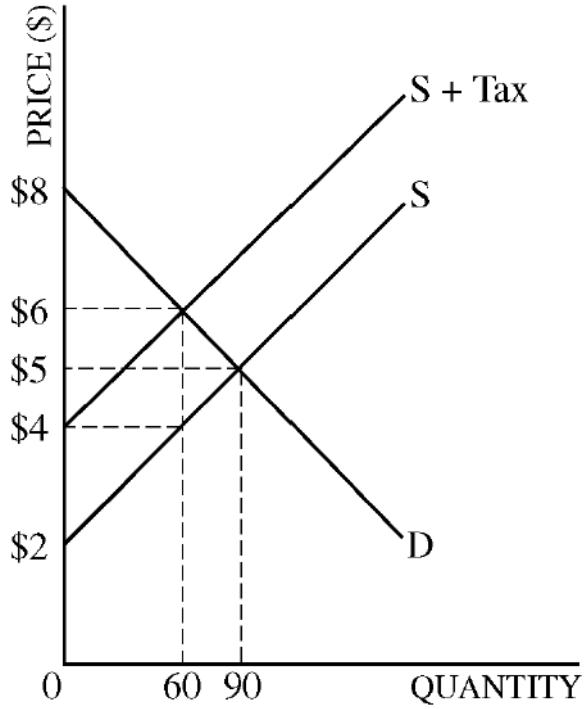
e. Assume the price decreases from $20 to $12.

i. Calculate the price elasticity of demand. Show your work.

ii. In this price range, is demand perfectly elastic, relatively elastic, unit elastic, relatively inelastic, or perfectly inelastic?

9. Respond to all parts of the question.

2.8 government intervention FRQ



The graph above illustrates the market for calculators. S denotes the current supply curve, and D denotes the demand curve.

a. Calculate the producer surplus before the tax.

b. Now assume a per-unit tax of $2 is imposed whose impact is shown in the graph above.

   i. Calculate the amount of tax revenue.

   ii. What is the after-tax price that the sellers now keep?

   iii. Calculate the producer surplus after the tax.

c. Is the demand price elastic, inelastic, or unit elastic between the prices of $5 and $6 ? Explain.

d. Assuming no externalities, how does the tax affect allocative efficiency? Explain.

10. Respond to all parts of the question.

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Assume the market for good Y is in equilibrium.

(a) Draw a correctly labeled demand and supply graph for good Y. Label the equilibrium price P_e and the equilibrium quantity Q_e.

2.8 government intervention FRQ

(b) Assume the government imposes a per-unit tax on good Y. On your graph in part (a), show each of the following after the tax has been implemented.

(i) The equilibrium price labeled P_N and the equilibrium quantity labeled Q_N

(ii) The area representing the change in consumer surplus, shaded completely

(c) Will the price paid by consumers increase by the same amount as the tax? Explain.

(d) Will the loss in consumer and producer surplus be greater than, less than, or equal to the tax revenue collected by the government? Explain.

11. Respond to all parts of the question.

The market for apartments in the town of Westville is currently in equilibrium, and consumers are renting 700 apartments at a price of $1,000 per apartment.

a. Assuming that the supply of apartments is perfectly inelastic, draw a correctly labeled graph for the apartment market and label the equilibrium price and quantity using the values above.

b. Now assume the government imposes a price (rent) ceiling of $800 for an apartment. On your graph in part (a), show the following.

   i. The price ceiling at $800

   ii. The quantity demanded, labeled as Q_d

   iii. The quantity supplied, labeled as Q_s

c. On your graph, shade completely the area representing the consumer surplus after the imposition of the price ceiling.

d. Calculate the dollar value of the producer surplus after the imposition of the price ceiling. Show your work.

e. Explain why the imposition of the price ceiling does not result in a deadweight loss.

12. Respond to all parts of the question.

---

13. Respond to all parts of the question.

Directions: You are advised to spend the first 10 minutes reading all of the questions and planning your answers. You will then have 50 minutes to answer all three of the following questions. You may begin writing your responses before the reading period is over. It is suggested that you spend approximately half your time on the first question and divide the remaining time equally between the next two questions.Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. Use a pen with black or dark blue ink.

Assume milk is produced and sold in a perfectly competitive market. To help milk producers, the government imposes an effective price floor on milk. Assume milk producers continue to sell milk only to households after the imposition of the price floor.

1. Draw a correctly labeled graph for the milk market and show the price floor, labeled P_F , and the quantity of milk

---

2.8 government intervention FRQ

purchased by households, labeled Q_F.

2. Is the market for milk allocatively efficient after the price floor is imposed? Explain.

3. On your graph from part (a), shade the area that represents total producer surplus after the price floor is imposed.

4. Assume the absolute value of price elasticity of demand for milk is 2 throughout the price range. Would total consumer spending on milk increase, decrease, not change, or be indeterminate after the imposition of the price floor? Explain.

5. Assume the price floor in the milk market caused the equilibrium price of eggs, a related consumption good, to increase.

    1. Draw a correctly labeled graph of the egg market, and show the effect of the price floor in the milk market on the egg market.

    2. What economic term describes the relationship between milk and egg?

---

a. The table below gives the quantity of good X demanded and supplied at various prices.

Price (dollars)	Quantity Demanded (units)	Quantity Supplied (units)
30	1	3
20	3	3
10	4	3

    i. Is the demand for good X relatively elastic, relatively inelastic, unit elastic, perfectly elastic, or perfectly inelastic when the price decreases from $30 to $20 ? Explain.

    ii. Is the supply of good X relatively elastic, relatively inelastic, unit elastic, perfectly elastic, or perfectly inelastic when the price decreases from $30 to $20 ? Explain.

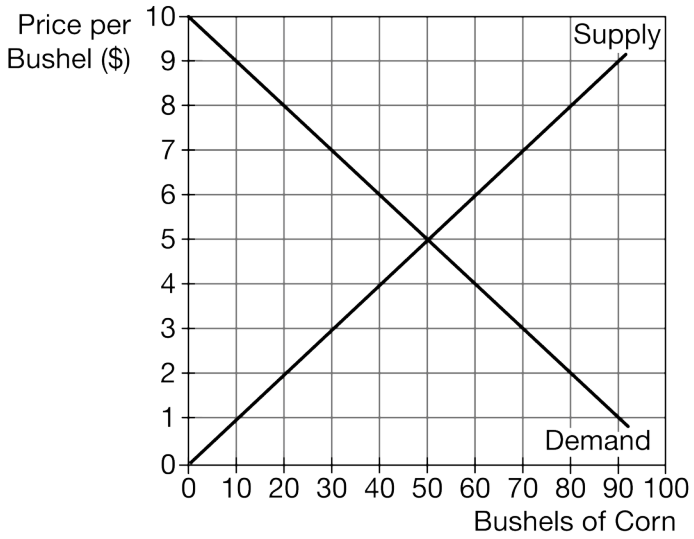
    iii. If a per-unit tax is imposed on good X, how is the burden of the tax distributed between the buyers and sellers of good X?

b. Assume that the income elasticity of demand for good Y is –2. Using a correctly labeled graph of the market for good Y, show the effect of a significant increase in income on the equilibrium price of good Y in the short run.

14. Respond to all parts of the question.

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

2.8 government intervention FRQ



The diagram above shows the market for corn in the country of Microland. Corn is produced and sold in a constant-cost, perfectly competitive market.

(a) Calculate the total revenue earned by corn farmers at the market equilibrium price. Show your work.

(b) In an attempt to assist corn farmers in Microland, the government sets a $7 price floor on corn.

(i) How many bushels of corn will be exchanged at the price floor?
(ii) Calculate the deadweight loss associated with the price floor. Show your work.
(iii) Assume the government agrees to buy the unsold quantity at $7. Calculate the producer surplus. Show your work.
(iv) Assume the price floor and the government buying program remain in effect. In addition, assume the demand for corn does not change. In the long run, will the quantity of corn purchased by the government increase, decrease, or remain the same? Explain.

15. Respond to all parts of the question: